

As a practical illustration we consider a simulated dataset generated as described in Section ???. In this simulated example there are $N = 700$ subjects, randomized 1:1, where the underlying (marginal) hazard ratio for the harm population H is $\theta^\dagger(H) = 2.5$, and for the complement H^c , the (marginal) hazard ratio is $\theta^\dagger(H^c) = 0.647$. The overall (marginal) hazard ratio is 0.71. Subjects' outcomes are generated from a Weibull regression model with 5 covariates, and analysis is based on those 5 covariates in addition to 2 “noise” covariates. The underlying proportion of subjects in the harm population H is $\approx 13\%$. Here the actual number of subjects in the H subgroup is 101 and the number of subjects in the complement equals 599.

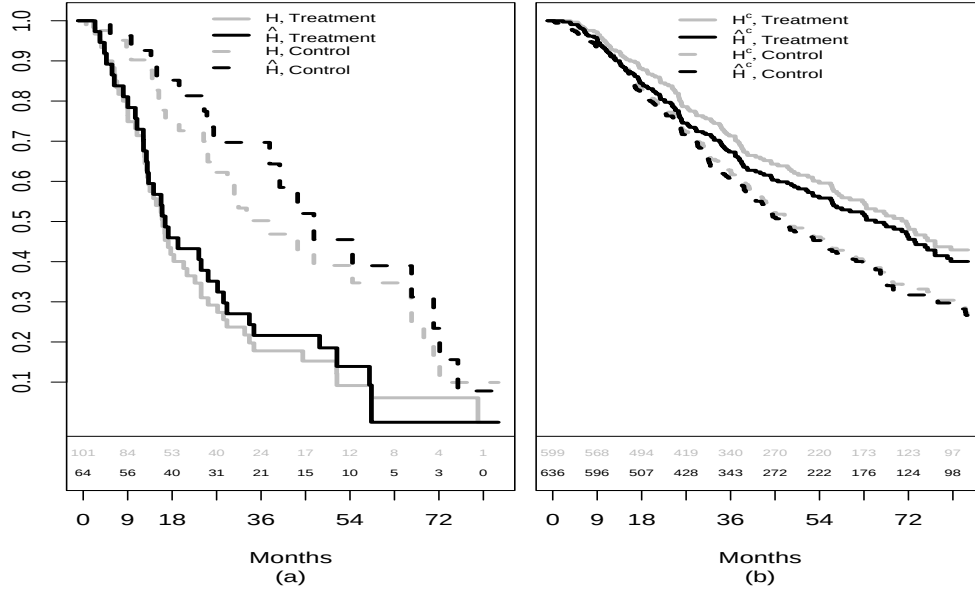


Fig. 1 Simulated data example with $\approx 44\%$ censoring: (a) Kaplan-Meier (KM) curves for the true H (H) subgroup for the treatment (solid grey) and control (dashed grey) arms, along with the KM curves for the estimated H ($\hat{H}$) subgroup treatment (solid black) and control (dashed black); (b) KM curves for the true complement H^c (H^c) subgroup for the treatment (solid grey) and control (dashed grey) arms, along with the KM curves for the estimated H^c ($\hat{H}^c$) subgroup treatment (solid black) and control (dashed black). The displayed risk sets correspond to the treatment and control arms combined for the true and estimated subgroups (H and $\hat{H}$ in (a); H^c and $\hat{H}^c$ in (b)).

Applying our novel procedure the estimated subgroup $\hat{H}$ consists of 64 subjects and thus 636 in the estimated complement $\hat{H}^c$. The number of subjects in H that are included in $\hat{H}$ is 52 (i.e., $|H \cap \hat{H}| = 52$): Of the 101 subjects in H , $52/101 = 51.5\%$ are correctly classified and of the 64 subjects in $\hat{H}$, $52/64 = 81.2\%$ are correctly classified. For the complement, the number of subjects in H^c that are included in $\hat{H}^c$ is 587: Of the 599 subjects in H^c , $587/599 = 98\%$ are correctly classified and of the 636 subjects in $\hat{H}^c$, $587/636 = 92.3\%$ are correctly classified.

In other words, $\hat{H}^c$ represents 98% of H^c , and for estimation (Cox and Kaplan-Meier) based on $\hat{H}^c$: 92.3% are H^c subjects with (marginal) hazard ratio $\theta^\dagger(H^c) = 0.647$, and 7.7% are H subjects with (marginal) hazard ratio $\theta^\dagger(H) = 2.5$.

Figure 1 displays the Kaplan-Meier curves based on the true (known a-priori) subgroups and the estimated subgroups (Figures 1(a) and 1(b), for H and H^c , respectively). The Kaplan-Meier curves for the treatment and control arms for the actual subgroups (H , known a-priori) are generally well characterized by the estimated subgroups.

Table 1 provides the Cox estimate for the overall population (ITT) as well as the estimates based on the true (known a-priori), and estimated subgroups, $\hat{H}$ and $\hat{H}^c$. The ITT Cox estimate is 0.84 (95% CI=0.69, 1.02), which is statistically insignificant and highlights the practical implications. For characterizing H , recall that the underlying (marginal) hazard ratio is $\theta^\dagger(H) = 2.5$. If the true H is known (a-priori), the Cox estimate is 2.36 (95% CI=1.47, 3.78), and for the estimated H , $\hat{H}$, the Cox estimate is 3.08 (95% CI=1.6, 5.9). By the nature of the FS procedure we expect Cox model estimates based on an estimated subgroup to be upwardly

Table 1 Simulated data example: Cox hazard ratio (hr) estimates for the ITT population and subgroups H and H^c . Cox model estimates are based on subgroups: H true (knowing the actual subgroup, a-priori); the estimated subgroup $\hat{H}$; and the bootstrap (500 resamples) bias-correction to $\hat{H}$ estimates, denoted $\hat{H}_{bc}$. Estimates for the complement H^c are defined analogously. The number of subjects in each population (# Subjects) and the number of subjects correctly classified (# Correct) in the true subgroups H and H^c , respectively, are listed.

	HR Estimate	Lower	Upper	# Subjects	# Correct
ITT					
ITT	0.838	0.687	1.021	700	.
H subgroup estimates					
H true	2.358	1.471	3.781	101	.
$\hat{H}$	3.077	1.604	5.904	64	52
$\hat{H}_{bc}$	2.248	1.436	3.519	64	52
H-complement subgroup estimates					
H^c true	0.672	0.537	0.841	599	.
$\hat{H}^c$	0.733	0.592	0.908	636	587
$\hat{H}_{bc}^c$	0.743	0.592	0.932	636	587

biased for H . We therefore describe a bootstrap bias-correction, which in this case, the bias-corrected Cox estimate denoted by $\hat{H}_{bc}$ is 2.25 (95% CI=1.44, 3.52) which we see provides a more “tempered” estimate and is quite similar to the estimates based on the ideal true H in this simulated example. In addition, notice that the CIs for each estimate cover the underlying (marginal) respective hazard ratios $\theta^\dagger(H) = 2.5$ and $\theta^\dagger(H^c) = 0.65$ for H and H^c , respectively. The CIs for $\hat{H}$ and $\hat{H}_{bc}$ also cover the estimate based on H true = 2.36; and similarly for the complement where H^c true = 0.67. We remark that this simulated example is representative of the results observed in our simulation setting described in Section ??.